Reproducible Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Reproducible Experiments

> Prepare Reproducible Experiments: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Record enough detail to reproduce a run

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a declared score threshold including equality

Reference example, not the only acceptable wording: Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Look for: Fit mean and median constant baselines on **training** targets

Reference example, not the only acceptable wording: Teacher explanation: Mean 6; median 5.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

Concept to explain

A result is reproducible only when another builder can recreate the data, split, code, settings, **seed**, and metric calculation.

Term	Meaning
reproducibility	ability to recreate a result
data version	identified dataset state
random seed	value controlling repeatable randomness
experiment record	complete inputs, settings, and outcomes

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on experimental records and **seed**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

The model **version**, split, metric, data provenance and development choices.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Apply the connected idea: Versioned experiments

Teach Week 28 as a two-session investigation using fictional data. A fictional activity scoring experiment compares two weight settings on the same development records.

Save code **version**, **input version**, weights, tie rule and **output**s. A label such as v2 is useful only when it refers to an identifiable saved state.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

To avoid changing expectations simply to match a bug.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can explain reproducible experiments using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Reproducible Experiments using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Reproduce a peer's run from an experiment card.

MODEL IDEA A result is reproducible only when another builder can recreate the data, split, code, settings, **seed**, and metric calculation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Record data=v2, split=grouped-s1, code=commit A7, **seed**=42. Record pre**process**ing, features, model type, settings, and library **version**s.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Record data=v2, split=grouped-s1, code=commit A7, **seed**=42.

Step 2

STEP 2 Record pre**process**ing, features, model type, settings, and library **version**s.

Step 3

STEP 3 Peer reruns and gets 16/20 correct, matching **80%** accuracy.

Step 4

STEP 4 A mismatch triggers environment and data comparison—not silent replacement.

CHECK Verify the denominator, unit, split, class definition, and **data version** before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

The teaching case

Specify **reproducibility** records.

Exact explanation to model

Why the result follows

Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

Explain the added worked example

Explain the added worked example · 1

A **seed** can help repeat a random split or initialization, but it is not a magic quality stamp. Save **data version**, split membership, code **version**, settings, pre**process**ing, **seed** and metric definition.

Some software operations remain nondeterministic, so document tolerances and environment too.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines experimental records and **seed**s.

A team declares **seed**s 11, 22, 33 before running. Validation accuracies are 14/20, 16/20, 12/20:70, 80, 60 percent. Mean accuracy is 70 percent, with a 20-percentage-point range.

Reporting only **seed** 22 as the model result hides the declared experiment.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Seed	An initial value controlling a pseudo-random process
Run record	A saved account of inputs, settings and outputs
Reproducibility	Ability to repeat a specified procedure
Cherry-picking	Reporting favorable results while hiding relevant others

Apply the connected idea: Versioned experiments

A fictional activity scoring experiment compares two weight settings on the same development records. Save code **version**, **input version**, weights, tie rule and **output**s.

A label such as v2 is useful only when it refers to an identifiable saved state.

The ranking changes because the preference weights changed. That is a policy comparison, not proof that v2 is more accurate.

Keep the same data and constraints when comparing weights, and record why a policy better fits the stated goal.

No. The data change is another possible explanation. Exit: No. They now informed development. Report that use and seek fresh cases for a new independent check.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Experimental records and seeds at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Before running	Declare 11 22 33	Save data and settings
Observed results	70 80 60 percent	Keep all three runs
Summary	Mean 70 percent	Range 20 points

The seed is part of the procedure, not evidence that a split represents the target population. Reproducing leakage perfectly reproduces a mistake. Check group separation and legitimate features before celebrating identical outputs.

A file contains only seed 22 and 80 percent. Name four missing items. Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Reproducible Experiments

Record variation across declared **seed**s instead of selecting the prettiest run.

The facts we will use

Supplied facts

Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/26

Reproducible Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep run identities

Record seeds4, 5, 6 with 6/10, 8/10, 7/10.



2. Summarise variation

Mean accuracy is 70%; range is 60% to80%.



3. Preserve context

Record data version, split, code, settings and seeds, not just the winning80%.

CHANGE ONE THING

The report deletes the 60% run.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep run identities	Record seeds4, 5, 6 with 6/10, 8/10, 7/10.
2. Summarise variation	Mean accuracy is 70%; range is 60% to80%.
3. Preserve context	Record data version, split, code, settings and seeds, not just the winning80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Seeds help reproduction but do not necessarily remove hardware or library nondeterminism;

report the actual environment.

Change one thing in the pictured case

Change one thing

Change: The report deletes the **60%** run.

Predicted result

Prediction: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Why the result follows

Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Reproducible Experiments, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

2. State the decision to explain

Write the question to explain: Specify **reproducibility** records.

3. Reveal the worked reasoning

Reveal after **prediction**s: Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

Hold back the independent task

Keep for an independent attempt: Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Examples: **data version**, split IDs, model settings, code **version**, metric denominator and full planned-run results.

Exit response: Record the failed run, cause and any retry; disclose which runs contributed to summaries and do not invent its score.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Use the prerequisite question: What must accompany a reported validation score? Check the explanation before proceeding.
7 to 18 minutes	Read the opening and model the worked evidence. Ask students to identify seed in context.
18 to 32 minutes	Complete W 1 and W 2 with support: summarize all runs; improve a record. Require the evidence, not only the final response.
32 to 45 minutes	Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.
Session 2 start to 8 minutes	Retrieve the distinction between seed and run record. Invite a counterexample to an overconfident claim.
8 to 20 minutes	Discuss the application and complete W 3: Does identical output prove a valid experiment?
20 to 35 minutes	Collect W4 independently before giving hints. Assess these outcomes: Record enough detail to reproduce a run; Distinguish reproducibility from validity. Record any support used.
35 to 45 minutes	Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: A file contains only **seed** 22 and 80 percent. Name four missing items.

Reasoning to listen for

Teacher check: Teacher explanation: Examples: **data version**, split IDs, model settings, code **version**, metric denominator and full planned-run results.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Experimental records and **seed**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Versioned experiments

Target: Record a reproducible experiment **version**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A has fun 3 and learning 2. Score it under weights (2,1) and (1,2).

Reasoning to listen for

Teacher check: Scores 8 and 7.

Individual check: For **Version**ed experiments, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: Once **holdout** results guide another change, those cases have become development **evidence**. A later independent evaluation needs new reserved cases. Small **holdout**s still provide limited **evidence**.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

The three individual run results, **seed** list, 60–**80%** range and shared evaluation conditions, so variability is visible.

Changed-case check: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Changed-case reasoning: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Reproducible Experiments investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Writing down the final accuracy makes an experiment reproducible.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

A stronger claim makes a stronger project: Record the failed run, cause and any retry; disclose which runs contributed to summaries and do not invent its score.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Mean 27/36=75 percent; range(10-8)/12=1/6 about 16.7 percentage points. Retain all three runs.

Check these misconceptions

A successful run proves the conclusion: Once **holdout** results guide another change, those cases have become development **evidence**. A later independent evaluation needs new reserved cases.

Small **holdout**s still provide limited **evidence**.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: E (4,2), F (1,5) gives v1 10 versus 7 and v2 8 versus 11.

Require a complete run log and a statement of what is held fixed.

A convenient shortcut preserves meaning: Overwriting a file called latest removes the ability to reproduce earlier results unless **version**s are retained elsewhere. Save identifiable **version**s and a run log.

Avoid editing a reserved final test after seeing its **output** to make the chosen **version** look better.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Reproducible Experiments questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

Independent question

Independent question · withhold the key

Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

Expected independent answer

Teacher key · Expected independent answer

Record **data version**, split or **seed**, features, model settings, code **version** and metric. Since multiple factors changed, the score difference cannot be attributed to one factor.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **random seed** alone records every source of experimental variation.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Record dataset and label **version**s, identity groups and split IDs, code/dependency **version**s, pre**process**ing, features, model settings, **seed**s, metric definitions and raw **prediction**s.

Same **seed** cannot compensate for different data or procedures.

Explanations for the additional workbook examples

W1 Summarize all runs

Workbook question · W1

Calculate the worked mean and range in percentage points.

Teacher reasoning · W1

Teacher explanation: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

W2 Improve a record

Workbook question · W2

A file contains only **seed** 22 and 80 percent. Name four missing items.

Teacher reasoning · W2

Teacher explanation: Examples: **data version**, split IDs, model settings, code **version**, metric denominator and full planned-run results.

W3 Separate two ideas

Workbook question · W3

Does identical **output** prove a valid experiment?

Teacher reasoning · W3

Teacher explanation: No. A deterministic experiment can still contain leakage, invalid labels or an unrepresentative split.

W4 Apply independently

Workbook question · W4

Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

Teacher reasoning · W4

Teacher explanation: Mean 27/36=75 percent; range(10-8)/12=1/6 about 16.7 percentage points. Retain all three runs.

W5 Handle a failed run

Workbook question · W5

One planned **seed** fails before **prediction**. How should it appear in the record?

Teacher reasoning · W5

Teacher explanation: Record the failed run, cause and any retry; disclose which runs contributed to summaries and do not invent its score.

Versioned experiments — W1 Record a run

List five items needed to reproduce a scoring run.

Code **version**, **input version**, **configuration** weights, tie/eligibility rules and **output**s; also record date and method as useful context.

Versioned experiments — W2 Compute a comparison

A has fun 3 and learning 2. Score it under weights (2,1) and (1,2).

Scores 8 and 7.

The weight pair is (fun weight, learning weight). Under (2,1), the score is 2*3 + 1*2 = 8. Under (1,2), it is 1*3 + 2*2 = 7. The same ratings were kept fixed; the difference comes from changing the preference policy.

Versioned experiments — W3 Repair an experiment

A learner changes weights, **input** ratings and the time limit together. Why is attribution difficult?

Multiple factors changed. Compare one deliberate change while holding the others fixed, or explicitly describe a multi-change experiment without attributing it to one cause.

Versioned experiments — W4 Design independently

Plan two **version**s for C (fun 5, learning 1) and D (fun 2, learning 5), using weights (2,1) and (1,2). Record expected scores, ranks, fixed data and a separate final-case plan.

v1 C 11, D 9 selects C; v2 C 7, D 12 selects D. Preserve the same records and other rules; reserve unseen cases until the policy is fixed.

Versioned experiments — W5 Protect evaluation

> You revise the policy after reading final-case failures. Are those cases still untouched **holdout evidence**?

No. They now informed development. Report that use and seek fresh cases for a new independent check.

Assess independent transfer

> **Independent question · withhold the key**
>
> Selected case: Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

Selected case answer

> **Teacher key · Selected case answer**
>
> Teacher explanation: Mean 27/36=75 percent; range(10-8)/12=1/6 about 16.7 percentage points. Retain all three runs.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Reproducible Experiments to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 27: Over**fitting**. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **seed** and **run record** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Mean 27/36=75 percent; range(10-8)/12=1/6 about 16.7 percentage points. Retain all three runs. Fresh retry accuracies 50, 60, 70 percent: mean 60 percent, range 20 percentage points.

Require all declared runs and their denominators. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Distribution shift. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Readiness and reassessment for Versioned experiments

Provide a run-log table. Extension: rerun a saved **version** in a fresh folder and compare exact **output** values.

Independent W4 **evidence**: v1 C 11, D 9 selects C; v2 C 7, D 12 selects D. Preserve the same records and other rules; reserve unseen cases until the policy is fixed.

Record whether the learner needed no help, a hint or explanation. Fresh retry: E (4,2), F (1,5) gives v1 10 versus 7 and v2 8 versus 11. Require a complete run log and a statement of what is held fixed.

Plan the delayed return

Suggested return points in this level: ai-4/27, ai-4/29, ai-4/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Reproducible Experiments lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Record enough detail to reproduce a run

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace **version** ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: A file contains only **seed** 22 and 80 percent. Name four missing items.

Supported practice answer

Reasoned practice answer: Teacher explanation: Examples: **data version**, split IDs, model settings, code **version**, metric denominator and full planned-run results.

Check every learner before discussion: Use the worked case for Experimental records and **seed**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The report deletes the **60%** run. Require a **prediction** and a reason before comparing. Expected result: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when

> this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

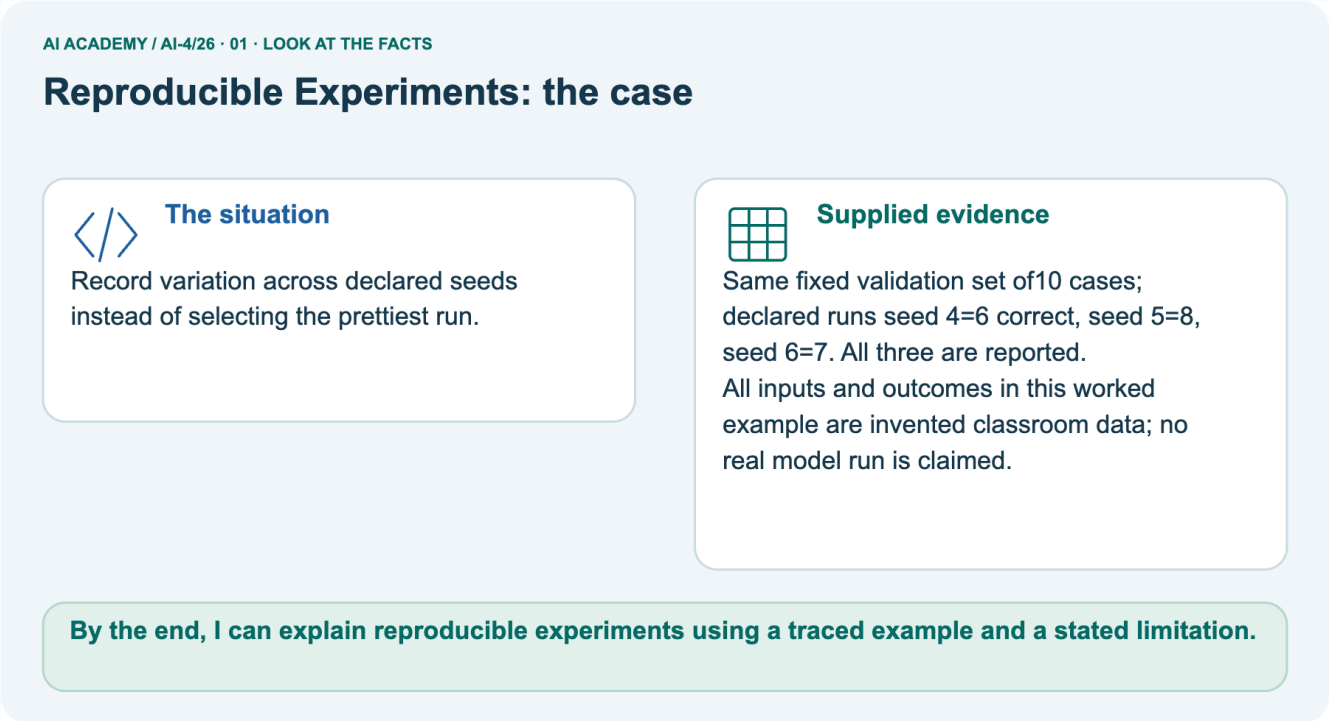
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Reproducible Experiments graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Reproducible Experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Record variation across declared **seed**s instead of selecting the prettiest run. Supplied facts: Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

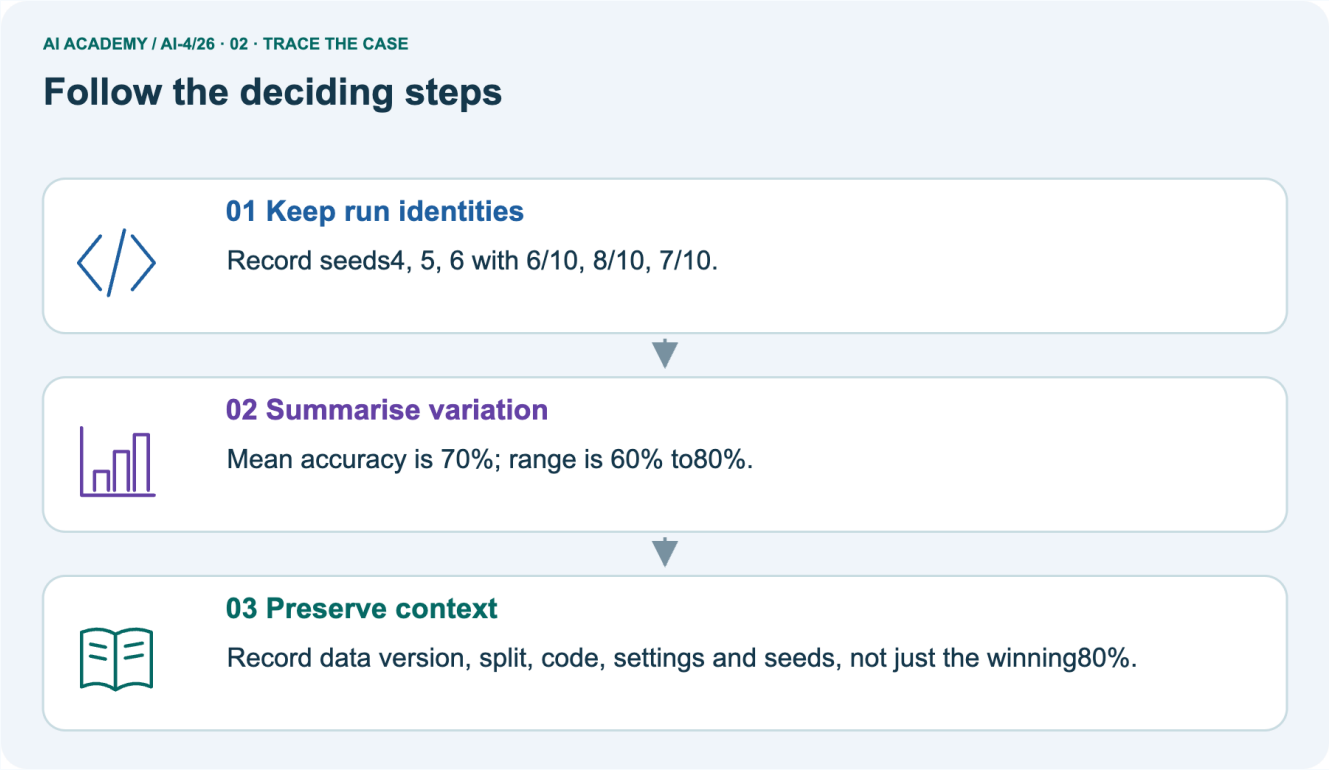
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep run identities: Record **seed**s4, 5, 6 with 6/10, 8/10, 7/10. → Summarise variation: Mean accuracy is **70%**; range is **60%** to**80%**. → Preserve context: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. All declared seeds



Follow the labelled picture with the matching explanation. Open the vector diagram

All declared **seed**s. Units: accuracy percent. **seed** 4: 60. **seed** 5: 80. **seed** 6: 70.

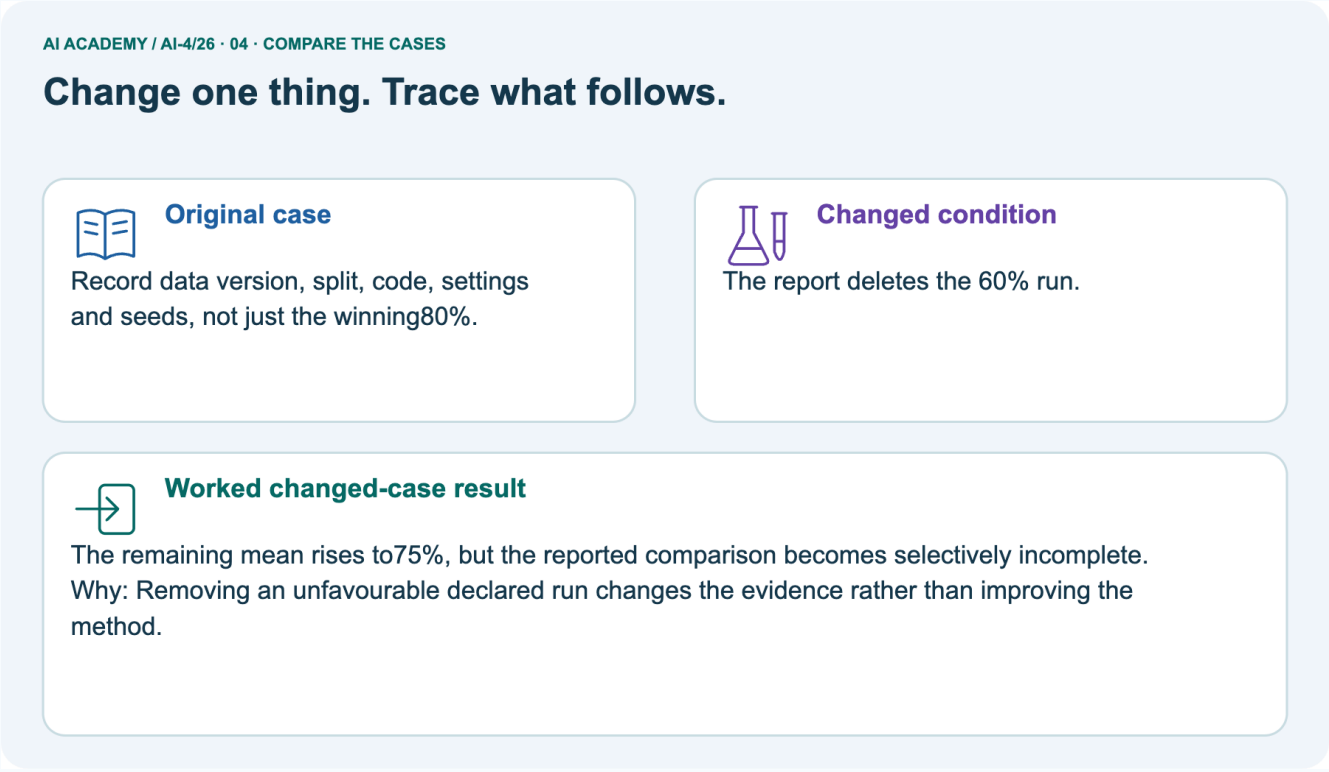
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Changed condition: The report deletes the **60%** run.

Expected result: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

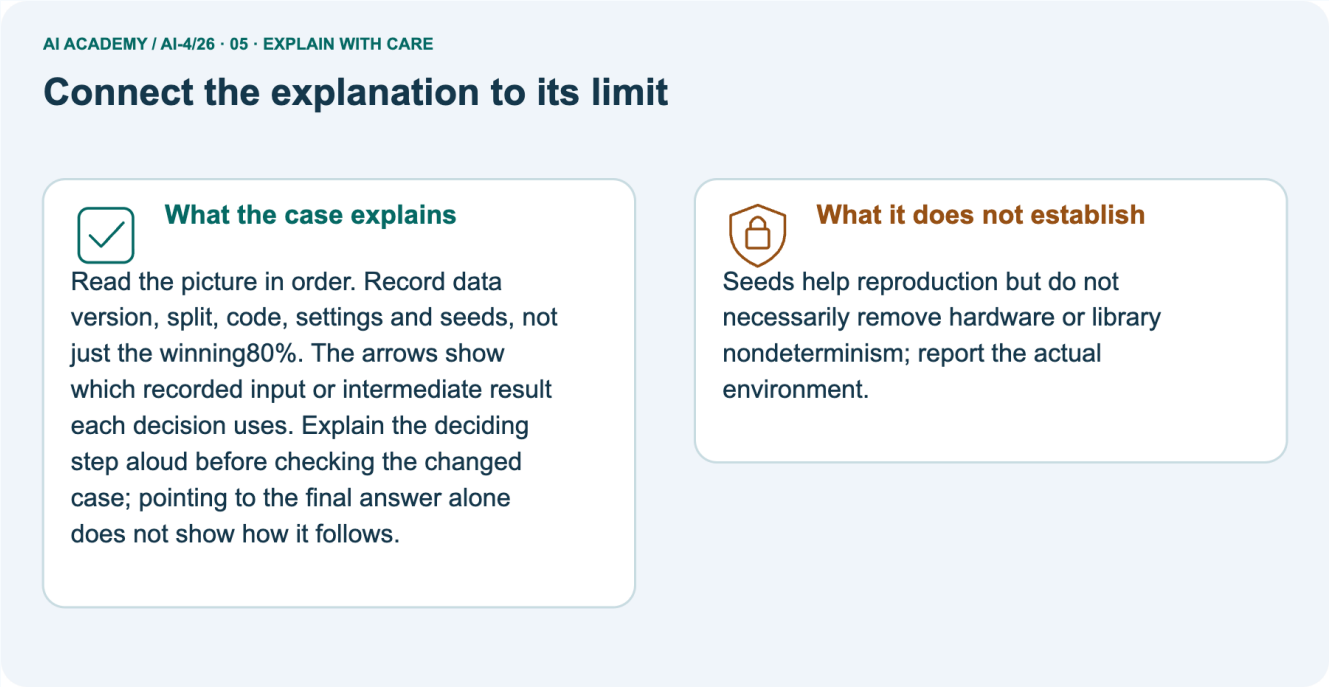
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

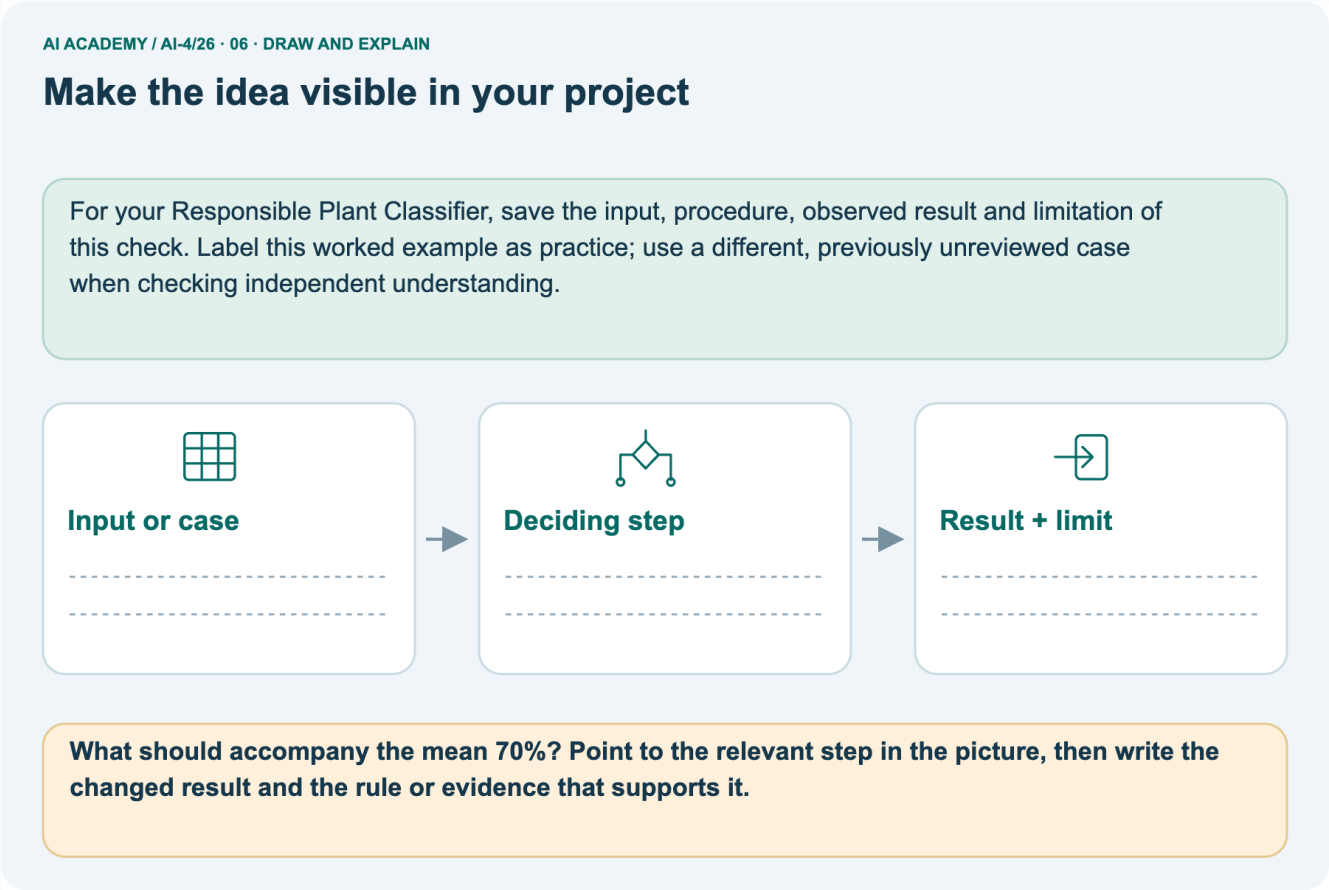
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The three individual run results, **seed** list, 60–**80%** range and shared evaluation conditions, so variability is visible.

Changed-case check: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.